

## PROGRAMMING FOR PROBLEM SOLVING CS201

TIME ALLOTTED: 3 HOURS

FULL MARKS: 70

*The figures in the margin indicate full marks.*

*Candidates are required to give their answers in their own words as far as practicable*

### GROUP – A (Multiple Choice Type Questions)

1. Answer any **ten** from the following, choosing the correct alternative of each question: **10×1=10**

| SL    | Question                                                                                                                                                      | Marks | Co          | Blooms Taxonomy Level |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------|-----------------------|
| (i)   | Determine the 10's complement of 98352<br>a) 99999                      b) 10000<br>b) 1647                      d) 1648                                      | 1     | 1           | 3                     |
| (ii)  | Resolution of externally defined symbols is performed by<br>(a) linker                      (b) loader<br>(c) compiler                      (d) assembler     | 1     | IT<br>201.2 | 2                     |
| (iii) | The extended ASCII character set uses<br>(a) 7 bits                      (b) 8 bits<br>(c) 16 bits                      (d) None of these                     | 1     | IT<br>201.2 | 2                     |
| (iv)  | If the two strings are identical, then strcmp() function returns<br>(a) 0<br>(b) 1<br>(c) YES<br>(d) None of these                                            | 1     | IT<br>201.2 | 2                     |
| (v)   | Which of the following cannot be checked in a switch-case statement?<br>(a) char<br>(b) int<br>(c) float<br>(d) long                                          | 1     | IT<br>201.1 | 4                     |
| (vi)  | If a variable is declared before the main then<br>(a) it is a logical error<br>(b) it is a syntax error<br>(c) it is a runtime error<br>(d) none of the above | 1     | IT<br>201.1 | 2                     |
| (vii) | What will be the output of the following program segment?<br>int main() {<br>printf("value is = %d",(10++));<br>return 0; }<br>a) value is = 10               | 1     | 1,2         | 3                     |

- b) value is = 11  
c) value is = 0  
d) Error
- (viii) `char txt[20]="Hello World!\0";` 1 3,4 1  
How many bytes are allocated by the definition above?  
Assume the size of each character as 1 byte.  
a) 11  
b) 12  
c) 13  
d) 20
- (ix) What is the output of the following code snippet? 1 1,2 3
- ```
int main()
{
    int i = 0,
        j = 0;
    for (i = 0; i < 5; i++)
    {
        for (j = 0; j < 4; j++)
        {
            if (i > 1)
            {
                break;
            }
        }
        printf("Hi \n");
    }
}
```
- a) Hi is printed 5 times  
b) Hi is printed 9 times  
c) Hi is printed 7 times  
d) Hi is printed 4 times
- (x) What is the output of the following code snippet? 1 1,2 3
- ```
int main()
{
    int x=25;
    if (!!x)
        printf("%d",!x);
    else
        printf("%d",x);
    return 0;
}
```
- a) 0  
b) 25

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- c) 1  
d) Error
- (xi) In C, if you pass an array as an argument to a function, what actually gets passed? 1 IT 201.2 1  
 (a) Value of elements in array  
 (b) First element of the array  
 (c) Starting address of the first element of array  
 (d) Starting address of the last element of array
- (xii) If p is an integer pointer with a value 1000 in a 32-bit system, then what will the value of p + 5 be? 1 IT 201.2 2  
 (a) 1005  
 (b) 1010  
 (c) 1004  
 (d) None of these

**GROUP – B**  
(Short Answer Type Questions)  
(Answer any three of the following) 3 x 5 = 15

| SL | Question                                                                                                       | Marks | Co       | Blooms Taxonomy Level |
|----|----------------------------------------------------------------------------------------------------------------|-------|----------|-----------------------|
| 2. | Illustrate the difference between ‘call-by-value’ and ‘call-by-reference’ concepts in C with suitable example. | 5     | 3        | 2, 4                  |
| 3. | (i) What is the return type of calloc function? Compare <i>calloc</i> and <i>malloc</i> functions?             | 1+2   | 4        | 1, 2                  |
|    | (ii) Write require C code for declaring a 2D float array of <i>r</i> rows and <i>c</i> columns dynamically.    | 2     | 3,4      | 6                     |
| 4. | Write the C code to count number of words in a user given English sentence without using string.h header file. | 5     | IT 201.2 | 6                     |
| 5. | Discuss different ‘jump statements’ in C with example.                                                         | 5     | IT 201.2 | 2                     |
| 6. | Write a C program to implement the following pattern:<br>1<br>12<br>123<br>1234                                | 5     | 1        | 6                     |

**GROUP – C**  
(Long Answer Type Questions)  
(Answer any three of the following) 3 x 15 = 45

| SL | Question                                                      | Marks | Co       | Blooms Taxonomy Level |
|----|---------------------------------------------------------------|-------|----------|-----------------------|
| 7. | (i) Convert the following:<br>(a) (36) <sub>10</sub> to octal | 2     | IT 201.2 | 5                     |

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|     |                                                                                                                                                                                                                                                                                             |        |             |                      |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------|----------------------|
|     | (b) (32.625) <sub>10</sub> to binary                                                                                                                                                                                                                                                        | 2      | IT<br>201.2 | 5                    |
|     | (c) (2AB) <sub>16</sub> to decimal                                                                                                                                                                                                                                                          | 2      | IT<br>201.5 | 5                    |
|     | (b) (10110.001) <sub>2</sub> to octal                                                                                                                                                                                                                                                       | 2      | IT<br>201.2 | 5                    |
|     | (ii) Write a program that converts all lowercase characters in a given string to its equivalent uppercase character.                                                                                                                                                                        | 7      | 5           | Analyze,<br>Evaluate |
| 8.  | (i) What is a function? How can you create your own function? Explain with proper example.                                                                                                                                                                                                  | 2+3    | 3           |                      |
|     | (ii) “Intelligent use of function with Call by Reference makes it possible to return more than one value at a time from the called function to the calling function”. Justify the statement with proper example.                                                                            | 5      | 2,3         |                      |
|     | (iii) Write a C program to find out GCD of two numbers using recursion.                                                                                                                                                                                                                     | 5      |             |                      |
| 9.  | (i) Given an unsorted array <i>A</i> of size <i>N</i> of non-negative integers, find a continuous sub-array which adds to the given number (taken as input). Develop a C program to print the starting and end index of the sub-array if exists, otherwise print -1.                        | 5+5=10 | CO4         | Analyze              |
|     | (ii) Write a C program to check whether a number taken as input is prime or not.                                                                                                                                                                                                            | 5      | CO3         | Apply                |
| 10. | (i) Explain file and FILE in C language                                                                                                                                                                                                                                                     | 3      | 5           | 1                    |
|     | (ii) Write C code to open a file named input.txt for reading and writing (write only the line) with proper error checking. Name of the respective file must collect from command line.                                                                                                      | 4      | 5           | 6                    |
|     | (iii) “Passing the base address of an array than the entire array is advantageous” – Justify this statement with example.                                                                                                                                                                   | 3      | 4           | 5                    |
|     | (iv) Design a <i>structure</i> to specify data on students given below:<br>Roll Number, Name, Department, Course, and Year of admission.<br>Assume that there are not more than 450 students in college.<br>Write a program to print names of all students who joined in a particular year. | 5      | 4,5         | 6                    |
| 11. | (i) Explain the term ‘garbage value’ and how to deal with garbage values at the time of programming?                                                                                                                                                                                        | 5      | IT<br>201.1 | 1                    |
|     | (ii) What is the purpose of including header files in a C program explain with example.                                                                                                                                                                                                     | 5      | IT<br>201.3 | 2                    |
|     | How to return multiple values from a function explain with example.                                                                                                                                                                                                                         | 5      | IT<br>201.4 | 2                    |